State-Directed Payments in Medicaid Managed Care: A Staggered Difference-in-Differences Evaluation of Hospital Financial Effects

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Abstract

Background. State-Directed Payments (SDPs) under 42 CFR 438.6(c) have grown from a niche financing tool into the dominant supplemental-payment channel in Medicaid, with projected spending of approximately $110.2 billion annually as of August 2024 approvals. Despite their fiscal scale, no prior peer-reviewed quasi-experimental evaluation has estimated SDPs' effects on hospital finance, care volume, or Medicaid beneficiary access. Section 71116 of the One Big Beautiful Bill Act (OBBBA), enacted July 4, 2025, caps non-grandfathered SDPs at 100% of Medicare in Medicaid-expansion states and 110% in non-expansion states, making the policy question newly urgent.

Methods. We construct the first longitudinal panel of every SDP preprint approved by the Centers for Medicare & Medicaid Services (CMS) for rating periods starting on or after February 1, 2023 (n = 1,038 arrangements, 43 states plus Puerto Rico, $296.2B gross projected total computable). Our pre-specified primary design was a §71116 event study using the Medicare-referenced cap as plausibly exogenous federal-statute variation. Hand-coding of all 26 in-scope arrangements in the public CMS corpus revealed no binding dose variation — 9 of 26 arrangements were parseable and all 9 had dose = 0, while 21 of 26 CMS letters designate the arrangement as likely grandfathered under §71116(b) until 2028. We therefore pivot to a Callaway–Sant'Anna (2021) staggered difference-in-differences design on first-SDP-approval cohorts, with Sun-Abraham, Borusyak-Jaravel-Spiess, and Goodman-Bacon cross-checks and Rambachan-Roth HonestDiD sensitivity bounds. Hospital-year outcomes come from the Medicare Healthcare Cost Report Information System (HCRIS), 2011–2025. A net-of-provider-tax treatment variable following Baicker and Staiger (2005) is constructed from state-level priors (MACPAC / KFF) for 99.5% of dollar exposure.

Results. The pooled CS-DiD ATT on hospital operating margin is −0.006 percentage points (SE 0.008; 95% CI −0.022, +0.011), statistically indistinguishable from zero. Net-of-tax weighting yields −0.003 (SE 0.008). Borusyak-Jaravel-Spiess imputation returns a positive ATT of +0.021 (SE 0.002), likely reflecting extrapolation bias given the short post-period and thin cohort support; we lead with CS-DiD. HonestDiD bounds at M = 1.0 on event time 1 are (−0.052, +0.060), and a 200-draw random-dose permutation places the observed ATT comfortably inside the null envelope (−0.013, +0.008). Provider-class heterogeneity reveals a positive and statistically distinguishable ATT at academic-medical-center (AMC) professional services (+0.010; 95% CI +0.001, +0.020) and a positive but imprecise point estimate at inpatient hospital services (+0.007; −0.003, +0.017); outpatient hospital and nursing facility classes are null. Placebo outcomes — total operating revenue (+$16.7M; +$6.0M, +$27.3M) and non-Medicaid inpatient days (+545; +144, +947) — are not null, suggesting the observed volume effects reflect a general-volume channel rather than Medicaid-specific cost-shifting.

Conclusions. Pre-OBBBA SDPs, on average and at the pooled level, produced no detectable operating-margin effect on acute-care hospitals within 2–3 years of the first CMS-public approval. The credible signal is concentrated at AMC-professional arrangements, which typically flow to academically-affiliated teaching hospitals. The placebo-outcome pattern disciplines the interpretation: we do not claim a Medicaid-specific margin effect. With OBBBA §71116 caps now binding for non-grandfathered arrangements and 21 of 26 in-scope arrangements temporarily shielded by §71116(b) until 2028, the policy-relevant causal event study becomes fieldable around Q2 2027. Pending a CMS FOIA response for the unredacted Addendum Tables 1.A–8.A workbook, this paper establishes the data infrastructure and methodological template for that analysis while providing the first quasi-experimental null on the pre-OBBBA regime.

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1. Introduction

State-Directed Payments (SDPs) are the central, and fastest-growing, supplemental-payment channel in the American Medicaid program, and they are also the largest unevaluated category of Medicaid spending. Under the 2016 Medicaid managed-care final rule (42 CFR 438.6(c)), states may direct Medicaid managed-care organizations (MCOs) to make supplemental payments to providers under arrangements that CMS reviews and approves through a "preprint" process. The payments can take the form of uniform dollar or percentage rate increases, minimum or maximum fee schedules, or value-based purchasing contracts. In the eight years since the 2016 rule took effect, SDPs have quietly displaced the fragmented upper-payment-limit (UPL) and pass-through payment architecture that preceded them and grown into a $110-billion-per-year financing category that now dwarfs the residual UPL program and rivals the disproportionate share hospital (DSH) program in aggregate fiscal significance [@macpac2024sdp; @macpac2024hospitalpayments]. The Government Accountability Office estimated that SDP spending exceeded 10% of Medicaid managed-care spending in 2022 and was forecast to reach 15% within a few years; it found that evaluations existed for only 48 of 215 renewed SDP arrangements [@gao2023sdp]. Yates and coauthors' 2025 cross-section of all SDPs with rating periods starting between February 2023 and September 2024 documented $144.3 billion in projected aggregate spending, of which only $7.8 billion (5%) flowed through value-based contracts in HCP-LAN Categories 2 through 4 [@yates2025valuebased]. The mass of SDP dollars is lump-sum fee bumps to hospitals, academic medical centers, and nursing facilities.

The policy question is newly urgent because of Section 71116 of the One Big Beautiful Bill Act (OBBBA), enacted on July 4, 2025. Section 71116 caps non-grandfathered SDPs at 100% of Medicare in Medicaid-expansion states and 110% of Medicare in non-expansion states for rating periods beginning on or after the date of enactment, and it phases down grandfathered SDP payment levels by 10 percentage points per year beginning in 2028 [@cms2026obbba71116]. KFF's 2025 analysis identified roughly 29 of 42 managed-care states as likely to face meaningful revenue reductions [@kff2025reconciliationsdp]. Section 71116 is, in other words, the first federal statutory cap on a financing instrument that had previously been governed almost entirely by CMS's preprint-review discretion. The policy community therefore has a first-order need for a credible causal estimate of what SDPs were doing before the cap — both to anticipate what §71116 will undo and to discipline any eventual §71116 event-study analysis.

This paper takes up that question and produces a deliberately cautious causal estimate. The pre-registered primary design was the §71116 event study itself, treating the Medicare-referenced cap as plausibly exogenous federal-statute variation and constructing a state-provider-class dose variable from arrangement-level CMS Addendum Table 2.A "Provider Payment Analyses" data. Hand-coding of all 26 in-scope arrangements in the public CMS preprint corpus revealed that the dose variable has no identifying variation in currently-available data: 9 of 26 arrangements were successfully parsed from Table 2.A and all 9 have dose equal to zero (pre-cap payment level at or below the applicable cap), 21 of 26 CMS approval letters explicitly designate the arrangement as likely grandfathered under §71116(b), and the remaining 17 of 26 arrangements have no populated Table 2.A in the PDF-embedded addendum because states submit the Excel workbook separately to CMS and only sometimes merge a printed copy into the approval-package PDF. This is not an OCR failure — the PDFs are text-extractable — it is a structural feature of the CMS public-posting practice. We disclose this finding as the central methodological caveat of the paper and re-cast the §71116 design as a descriptive supplement that confirms the cap is not yet binding in public-record form.

The primary causal design of this paper is therefore the originally secondary design: a staggered Callaway-Sant'Anna difference-in-differences analysis of first CMS-public SDP approval on hospital operating margin, uncompensated-care ratio, total inpatient days, and Medicaid inpatient share, over 2011–2025 hospital-year HCRIS data. The treatment cohort g is the earliest CMS-approval year in a given state (or, in heterogeneity analyses, a given state $\times$ provider-class cell). The comparison group is the 13 HCRIS states with no CMS-public SDP approval in the panel window. We lay down four independent disciplines against the central threats in this design — endogenous adoption, short post-period, left-censored cohort timing, and the Baicker-Staiger "phantom transfer" problem. First, Sun-Abraham IW, Borusyak-Jaravel-Spiess imputation, and Goodman-Bacon decomposition as cross-check estimators [@sun2021estimating; @borusyak2024revisiting; @goodmanbacon2021did]. Second, Rambachan-Roth HonestDiD relative-magnitudes bounds at $M \in \{0.25, 0.5, 1.0, 1.5, 2.0\}$ [@rambachan2023credible]. Third, a 200-draw random-dose permutation null distribution. Fourth, a net-of-provider-tax sensitivity analysis over a 10%–35% recapture-rate span, following the Baicker-Staiger (2005) lesson that gross supplemental payments can be almost entirely recaptured through hold-harmless provider-tax loops [@baicker2005fiscal].

The paper's central empirical finding is a carefully-qualified null. The pooled CS-DiD ATT on operating margin is −0.006 (SE 0.008), statistically indistinguishable from zero, robust to net-of-tax weighting and to the HonestDiD bounds at M = 1.0. Borusyak-Jaravel-Spiess imputation delivers a markedly different point estimate (+0.021; SE 0.002), which we attribute primarily to extrapolation bias given the thin cohort structure and the short post-period. The signal that *does* survive is provider-class heterogeneity: the ATT at academic-medical-center-professional arrangements is +0.010 (95% CI +0.001, +0.020), the ATT at inpatient hospital services is +0.007 (−0.003, +0.017), and the ATTs at outpatient hospital and nursing facility arrangements are null. The two placebo outcomes — total operating revenue (not Medicaid-specific) and non-Medicaid inpatient days (mechanically non-Medicaid) — also reject the null, which disciplines our interpretation: we do not claim a Medicaid-specific margin effect. The most honest reading of the full-result package is that CMS-public SDP approvals are associated with a modest general-volume expansion at teaching hospitals and large urban safety-net systems, not with a Medicaid-cost-shifting channel.

This paper makes four contributions. First, we construct the first longitudinal, arrangement-level panel of CMS-public SDPs. All 1,038 approved preprints, with full parsed metadata, the net-of-tax treatment variable, the §71116 dose, and the state × provider-class first-approval cohorts, are documented in a reproducible pipeline from raw CMS HTML and PDFs to a clean panel (§3). Second, we contribute the first quasi-experimental null on the pre-OBBBA SDP regime, disciplined by HonestDiD, permutation, and placebo robustness. Third, we document the Baicker-Staiger "phantom transfer" limitation as a first-order measurement problem in the public CMS corpus — only 4 of 1,038 arrangements (2.0% of dollars) have arrangement-specific net-of-tax data — and report a sensitivity analysis over a realistic span of state-level recapture priors. Fourth, we establish the methodological template and data infrastructure for a §71116 event study that will become fieldable approximately Q2 2027, once HCRIS cost-report data for fiscal year 2026 are released and grandfathered arrangements begin their 2028 phase-down. A CMS FOIA request for the unredacted Addendum Tables 1.A–8.A workbook set has been filed; we discuss the follow-on paper that a successful FOIA response would unlock (§7).

The paper is organized as follows. Section 2 describes the institutional setting of SDPs, the CMS preprint process, and OBBBA §71116. Section 3 describes the data and the construction of the treatment and net-of-tax variables. Section 4 describes the estimators and robustness battery. Section 5 presents the results, leading with the CS-DiD pooled null on operating margin, the AMC-professional heterogeneity signal, and the placebo-outcome pattern, and concluding with the §71116 descriptive supplement. Section 6 discusses the findings in the context of the Medicaid-supplemental-payments and hospital-finance literatures, draws the policy implications for the OBBBA §71116 rollout, and catalogues limitations. Section 7 concludes.

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2. Background and Institutional Setting

2.1 The Medicaid financing stack and the rise of SDPs

Understanding SDPs requires placing them in the wider Medicaid hospital-payment stack. Base Medicaid payments to hospitals are set by state methodologies and, under managed care, are paid by MCOs out of risk-adjusted capitation revenue. Base payments are typically well below Medicare and commercial rates. Historically, the gap between base Medicaid rates and "reasonable" cost has been closed by a constellation of supplemental-payment mechanisms: Disproportionate Share Hospital (DSH) payments (authorized by §1923 of the Social Security Act), Upper Payment Limit (UPL) payments in fee-for-service Medicaid, and — until the 2016 rule — managed-care "pass-through" payments that replicated the UPL in the MCO environment [@macpac2021upl].

The 2016 Medicaid managed-care final rule explicitly prohibited pass-throughs and created two new mechanisms to replace them: state-directed payments (SDPs) under 42 CFR 438.6(c), and the value-based payment / minimum-fee-schedule path under 42 CFR 438.6(d). States quickly converged on SDPs as the dominant tool, with annual projected spending growing from approximately $25 billion in 2019 to $110 billion in MACPAC's October 2024 estimate [@macpac2024sdp]. The 2024 Medicaid managed-care rule (42 CFR Parts 438, 440, and 457) tightened reporting and transparency requirements but preserved SDPs as an instrument [@cms2024mcrule]. The GAO's

2023 audit found that CMS reviews SDPs against statutory requirements (actuarial soundness, permissible payment types, state-specific quality strategies) but does not systematically incorporate state-provided evaluation evidence when deciding on renewals [@gao2023sdp].

2.2 The CMS SDP preprint process and the public corpus

An SDP arrangement enters the public record when CMS approves a "preprint" — a standardized form in which the state describes the arrangement's scope, payment methodology, provider class, funding mechanism, and projected total computable (TC) dollar amount. Since early 2023, CMS has posted every approved preprint on medicaid.gov with metadata including state, provider class, approval date, rating-period dates, payment type (fee schedule or value-based), and review type (new, renewal, or amendment). The approval package PDF nominally includes Addendum Tables 1.A through 8.A with structured line-item data on payment methodology, IGT transferring entities, provider-tax arrangements, and arrangement-specific quality goals. In practice — as our hand-coding exercise documents (§3.5) — states submit the Excel workbook separately to CMS, and only sometimes merge a printed copy back into the approval-package PDF. The public corpus is therefore rich in narrative text and boilerplate but sparse in structured arrangement-level data.

A critical feature of the public corpus is that CMS does not post preprints approved before February 1, 2023. Pre-2023 arrangements, which include essentially all of the high-dollar hospital SDP arrangements in Louisiana, California, Illinois, Michigan, and other early-adopter states, are absent from the public record. They are obtainable only via CMS FOIA. The 2023 posting cutoff is therefore a structural data boundary, not a true policy-timing event. It concentrates observable first-approval cohorts into the 2023 calendar year (41 of 43 treated states in our panel) and creates a "single-cohort" staggered-DiD design that is more fragile than the full pre/post panel would be if pre-2023 data were publicly posted.

2.3 Provider classes, payment types, and dollar concentration

CMS records 11 provider classes in the approved-preprint metadata. In our panel, inpatient hospital service arrangements are the most numerous (n = 341) and, by a large margin, the most fiscally significant. Of the top-20 state × provider-class cells by gross projected TC, 19 are inpatient hospital services. The other four "in-scope for §71116" classes — outpatient hospital service (n = 50), nursing facility service (n = 79), and professional services at an academic medical center (n = 77) — are smaller in count but concentrated in a few states. The remaining seven classes (behavioral-health inpatient and outpatient, primary care, specialty physician, home-and-community-based / personal care, dental, and "other") account for about 35% of preprint counts but a smaller share of TC dollars. 894 of 1,038 arrangements are fee-schedule SDPs; 144 are value-based, consistent with Yates et al. (2025)'s estimate that only ⍰5% of SDP dollars flow through HCP-LAN Category 2–4 value-based contracts [@yates2025valuebased]. 737 of 1,038 are renewals, 172 are new arrangements, and 129 are amendments.

2.4 Net-of-tax mechanics: the Baicker-Staiger (2005) channel

A first-order institutional fact — under-appreciated in the health-services literature but pervasive in MACPAC and KFF reporting — is that most SDP financing runs through provider taxes and IGTs, not state general funds [@macpac2024sdp]. Under CMS regulations, a provider tax that redistributes substantially all revenue back to taxed providers is impermissible "hold-harmless" financing, but the 6% safe-harbor threshold (the maximum provider-tax rate that is presumed to satisfy the hold-harmless test) allows most state-hospital and state-nursing-facility arrangements to pass muster. KFF estimates provider-tax revenue funds roughly 18% of the state share of Medicaid spending nationally [@kff2024budgetsurvey]. For SDPs specifically, this matters because the net transfer to providers is the gross SDP minus the tax contribution; when the tax is calibrated to recapture most of the SDP revenue, the net transfer is a fraction of the headline number. This is exactly the expropriation channel that Baicker and Staiger (2005) identified for DSH: in states where fiscal institutions allowed most DSH dollars to be retained by public hospitals, Baicker and Staiger showed significant reductions in infant and pediatric mortality; in states where DSH dollars were expropriated for general budget purposes, the same nominal transfers had no mortality effect [@baicker2005fiscal]. We construct a net-of-provider-tax treatment variable in §3 as the primary operationalization of this lesson. We also disclose the measurement limitation: only 4 of 1,038 arrangements have arrangement-specific IGT data; the remaining 99.5% of dollar exposure is imputed from state-level priors.

2.5 OBBBA Section 71116: the federal cap

Section 71116 of OBBBA, effective July 4, 2025, caps SDP payment levels at 100% of Medicare in Medicaid-expansion states and 110% of Medicare in non-expansion states. The cap applies to the total payment level (base + SDP) in the provider's class, not to the SDP component alone. §71116 has three implementation features worth documenting explicitly:

1. Grandfathering under §71116(b): arrangements submitted or approved by July 4, 2025 with rating periods beginning within 180 days thereafter are "likely grandfathered" — payment levels freeze at current levels and phase down by 10 percentage points per year beginning in 2028. In the public corpus, 21 of 26 in-scope arrangements are so designated by CMS.
2. Rate basis: the cap is expressed as a percent of Medicare. States that benchmark SDPs to the "average commercial rate" (ACR) — which Marr et al. (2023) document as roughly 185% of Medicare on average for outpatient hospital services — face the steepest nominal cut if the ACR benchmark is not already below the cap [@marr2023hospital]. Approximately 10% of preprints in our panel reference ACR benchmarking.
3. Temporal bite: because grandfathering absorbs virtually all currently-approved arrangements, binding cap reductions do not begin until rating periods renewing without grandfathering eligibility or until 2028 (when the 10-pp/year phase-down starts for grandfathered arrangements).

A clean §71116 event study therefore requires HCRIS data covering rating periods that begin after July 4, 2025 in genuinely non-grandfathered arrangements. HCRIS cost-report filings for fiscal year 2026 are expected around mid-2027; the combination of the thin observable §71116 dose in the currently-public corpus and the HCRIS lag places the credible field date for a §71116 event study at approximately Q2 2027.

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3. Data

This section integrates the Agent 3 data-section draft with additions from the hand-coding pass. The primary data product is `data/clean/sdp_panel.parquet`, the master tidy panel of 1,038 CMS-approved SDP arrangements. All scripts reproduce deterministically from `data/raw/` to `data/clean/` via a single shell driver (`data/scripts/` in numerical order).

3.1 Primary data source: CMS approved SDP preprints

We construct a novel, arrangement-level panel of every SDP preprint approved by CMS for a state Medicaid managed-care rating period beginning on or after February 1, 2023. CMS began posting approved preprints centrally on medicaid.gov in early 2023 under 42 CFR 438.6(c). The published index carries standardized metadata for each arrangement: a unique SDP identifier of the form `{STATE}_{PAYMENT-TYPE}_{PROVIDER-CLASS}_{REVIEW-TYPE}_{RAT` the state, a narrative description including the projected TC dollar amount, the approval date, the effective date, the state rating period, the payment type, the review type, the approval period, the provider class, and a link to the combined PDF approval package.

We scraped the CMS preprint index on April 16, 2026, yielding 1,038 unique approved preprints from 43 states plus Puerto Rico spanning rating periods from October 1, 2020 to April 1, 2026. The scraping script (`data/scripts/01_scrape_sdp_preprints.py`) paginates the Solr-backed index via a deterministic GET interface at `limit=100` per page, caches each HTML page immutably, and downloads every referenced approval-package PDF. 1,035 of 1,038 PDFs downloaded successfully; three failed due to CMS listing-page HREF quirks (duplicate path prefix) and are flagged for manual retrieval.

We parse each PDF with `pdfplumber` (`data/scripts/02_parse_excel_addenda.py`) and extract text-level features: funding-mechanism indicator language (provider tax, IGT, CPE, state general fund, local tax, hold-harmless clause), non-federal-share-scoped funding flags constrained to the CMS preprint question on financing, percent-of-Medicare and percent-of-ACR benchmarking mentions, and validated dollar amounts.

An important constraint: CMS does not publish the structured Excel addendum workbooks (Tables 1.A–8.A) as separate files on the approved preprints index. They are embedded within the approval-package PDFs, frequently as flattened or scanned pages. OCR plus table-structure extraction at scale is out of Phase 3 scope. Our text-level features therefore provide only coarse indicators of funding mechanism and benchmarking. Specifically, the generic provider-tax / IGT indicator fires on CMS-template boilerplate that appears in every preprint (>99% positive rate) and carries little discriminating power; the non-federal-share-scoped flag is substantially more informative. Where informative text signals are absent, funding-mechanism classification defaults to state-level priors compiled from MACPAC (2024) and KFF (2024, 2025) [@macpac2024sdp; @kff2024budgetsurvey; @kff2025reconciliationsdp].

3.2 Treatment-variable construction

The paper's identification strategy, re-specified after the §71116 dose hand-coding revealed no identifying variation, has two layers. The primary strategy is a staggered Callaway-Sant'Anna DiD on first post-February-2023 SDP approval by state (and, in heterogeneity analyses, by state × provider-class). The supplementary strategy is the §71116 event study, now framed descriptively. We construct three treatment variables:

1. Net-of-provider-tax treatment (`data/scripts/03_build_net_of_tax_treatment.py`). Per the Baicker-Staiger (2005) lesson, the gross projected-TC amount overstates the true transfer when part of the non-federal share is recaptured through a provider tax or IGT hold-harmless loop. We combine arrangement-level text flags with state-level priors on non-federal-share composition to estimate an arrangement-specific recapture rate bounded above by the state's non-federal share. The resulting variable, `net_tc_usd = gross_tc_usd × (1 - recapture_rate_est)`, is the primary treatment dose for outcome weighting.

2. §71116 dose (`data/scripts/04_build_section_71116_dose.py` + `data/scripts/handcoding/build_71116_dose_handc` For each arrangement, `dose = (pre_cap_rate - min(pre_cap_rate, cap)) / pre_cap_rate`, where `cap = 100` in Medicaid-expansion states and `110` in non-expansion states. Pre-cap rate is the maximum across provider sub-classes of (base + SDP) as a percent of Medicare, extracted from Addendum Table 2.A "Provider Payment Analyses". Grandfathered arrangements under §71116(d) are flagged separately.

3. Staggered-DiD treatment cohorts (`data/scripts/05_build_csdid_treatment.py`). For each (state, provider-class) cell, we identify the first observed approval date. This yields 205 state × provider-class cohorts. Aggregating to the state level, 41 of 43 SDP states have g = 2023, 1 has g = 2024 (DC), and 1 has g = 2025 (CO). The 13 HCRIS states with no CMS-public SDP approval (AK, AL, AS, CT, GU, ID, ME, MP, MT, ND, SD, VI, WY) serve as the never-treated comparison group.

3.3 Outcome panel

Hospital-year outcomes come from HCRIS via the sibling DSH-LIUR paper's Phase 3 build (`papers/dsh-liur-threshold-rd/dat` which harmonizes 77,451 hospital-years from 2011 through 2025. We use the acute-hospital subsample (42,804 hospital-years) for the main analysis. Primary outcomes are:

1. Operating margin (winsorized at 1st/99th): (net patient revenue − operating expenses) / operating revenue.
2. Uncompensated-care ratio: uncompensated-care cost / operating expenses.
3. Total inpatient days.
4. Medicaid share of inpatient days.

Placebo outcomes are total operating revenue (a Medicaid-agnostic revenue metric) and non-Medicaid inpatient days (constructed as `total_ip_days × (1 - medicaid_share_days)`).

3.4 Summary coverage

The constructed panel covers 43 states plus Puerto Rico, 11 provider classes, rating periods 2020-Q4 through 2026-Q2, $296.2 billion gross projected TC across the panel, and $223.0 billion net-of-provider-tax, implying an average implied recapture rate of approximately 24.7%. Of the 1,038 arrangements, 547 are in the four §71116 in-scope provider classes and 26 have rating periods starting after July 4, 2025 and are non-grandfathered (the §71116 *active-treatment* sample). Annualized gross spending reconciles to the same order of magnitude as MACPAC's $110 B/yr 2024 estimate after accounting for multi-rating-period arrangements and overlap across years.

3.5 Hand-coding pass: what the public CMS corpus does (and does not) reveal

On 2026-04-16 the author conducted a systematic hand-coding pass of the 26 §71116-in-scope arrangements and the top-200 arrangements by gross TC (95% of dollar exposure). The objective was to (a) extract the §71116 dose variable from PDF-embedded Addendum Table 2.A and (b) extract arrangement-specific net-of-tax IGT and provider-tax data from Tables 4.A and 5.A.

The §71116 results are striking. Of 26 in-scope arrangements:

- 9 of 26 had parseable Table 2.A data. All 9 have dose = 0 (pre-cap rate at or below the applicable cap). Mean pre-cap rate across the 9 parsed arrangements = 80.5% of Medicare, with a maximum of 101.3% (still below the 100%/110% cap given the non-expansion-state designation). No arrangement in the public corpus has a positive binding dose.
- 21 of 26 are CMS-designated as "likely qualifying for the temporary grandfathering period in section 71116(b)" per the CMS approval letter text. Grandfathered arrangements are frozen at current payment levels and do not begin phase-down until 2028.
- 17 of 26 had no populated Table 2.A in the PDF: 14 lacked the heading entirely and 3 had the heading but blank row fields. This is not an OCR problem (the PDFs are text-extractable, with mean 40,000 characters per document); the structured Excel workbook is simply not merged back into the public approval-package PDF.

The net-of-tax hand-coding results are equally stark. Of the top-200 arrangements by gross TC:

- 4 of 200 have a populated Addendum Table 4.A with IGT transferring entities and dollar amounts (3 Louisiana inpatient-and-outpatient hospital arrangements and 1 Florida inpatient-and-outpatient hospital arrangement). Arrangement-specific coverage is $5.8 billion out of $296 billion gross (2.0%).
- 0 of 200 have a populated Addendum Table 5.A (provider tax).
- The remaining 196 top-200 and all 835 bottom arrangements retain the state-level MACPAC / KFF recapture prior, with a provenance tag (`state_prior_top200_reviewed` for the top-200 audited or `state_prior` for bottom-835 unreviewed).

The structural implication is clear: the CMS-public corpus is designed for regulatory transparency at the arrangement level for metadata and narrative, but not for empirical analysis at the arrangement level of structured payment details. A CMS FOIA request for the unredacted Addendum workbooks has been drafted (`data/foia-request-letter-cms.md`) and is pending user authorization to submit. Expected turnaround is 6–12 months.

The causal-design implication is that the §71116 event study is not currently fieldable with observable variation in the cap, and the net-of-tax treatment is effectively a state-level measure for 99.5% of dollar exposure. Both limitations motivate the §4 robustness strategy: HonestDiD bounds rather than point-level parallel-trends tests, a net-of-tax sensitivity over the full plausible 10%–35% recapture span, and a random-dose permutation null against which to benchmark the observed ATT.

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4. Methods

4.1 Design overview

We estimate the causal effect of SDP arrangements on hospital outcomes using the Callaway-Sant'Anna (2021) staggered DiD estimator of the group-time average treatment effect on the treated, ATT(g,t), aggregated to an overall post-period ATT and to event-time dynamic ATT(e) [@callaway2021did]. The treated units are HCRIS acute-care hospitals in states with a CMS-public SDP approval; the comparison units are HCRIS acute-care hospitals in the 13 states with no CMS-public SDP approval in the panel window. We use 2016 as the first pre-treatment year and 2025 as the last post-treatment year, yielding 7 pre-treatment years for the dominant g = 2023 cohort and 2–3 post-treatment years depending on HCRIS coverage.

4.2 Estimator: Callaway–Sant'Anna ATT(g,t)

For each treated cohort g ⍰ {2023, 2024, 2025} and each calendar year t, we estimate the canonical 2×2 group-time average treatment effect as

$$ATT(g, t) = \mathbb{E}[\Delta Y_{t,g-1} \mid G = g] - \mathbb{E}[\Delta Y_{t,g-1} \mid C_t = 1]$$

where $C_t = 1$ indicates the not-yet-treated or never-treated comparison group at year t, and $\Delta Y_{t, g-1} = Y_t - Y_{g-1}$. We use $g - 1$ as the reference period in both the pre-period placebo (leads) and post-period estimate (lags). Standard errors are the pointwise analytic SEs of the two-sample difference-of-means on first differences. We aggregate ATT(g,t) to event-time $e = t - g$ coefficients using cohort-size weights and to an overall post-treatment ATT using weighted averaging of all (g, t) with e ⍰ 0.

We implement three dose-response variants. First, a binary-treatment ATT. Second, a unit-weighted ATT where treated-side first differences are averaged using state-level gross TC as weights. Third, same weighting using state-level net-of-tax TC. Because cohort timing is effectively a single-date event (⍰95% of states have g = 2023), the binary and dose-weighted ATTs differ numerically only through the treated-side weighting; we report all three.

4.3 Cross-check estimators

We implement four heterogeneity-robust cross-checks on the primary outcome (operating margin winsorized):

1. Sun-Abraham IW event study via two-way fixed effects with cohort-indicator × event-time interactions, clustered at the state level [@sun2021estimating]. This is the direct event-study analog to the CS-DiD overall ATT.

2. Borusyak-Jaravel-Spiess (BJS) imputation estimator: fit unit + year fixed effects on the untreated sub-sample only; impute Y(0) for treated observations; compute ATT as the mean of observed-minus-imputed residuals [@borusyak2024revisiting]. BJS is efficient under homoskedastic errors and a parsimonious two-way FE untreated-potential-outcome model; it can extrapolate aggressively when the untreated sub-sample is small relative to the treated sub-sample.

3. Goodman-Bacon decomposition: decompose the TWFE estimate into 2×2 comparison weights (treated-vs-never, treated-vs-not-yet, late-vs-early, early-vs-already-treated) to quantify the share of identifying variation coming from "clean" vs. "contaminated" comparisons [@goodmanbacon2021did].

4. Rambachan-Roth HonestDiD relative-magnitudes bounds for M ⍰ {0.25, 0.5, 1.0, 1.5, 2.0} × the largest absolute pre-period coefficient, giving a formal robustness envelope for plausible parallel-trends deviations [@rambachan2023credible].

4.4 Placebo, permutation, and sensitivity

1. Placebo outcomes: we re-run the CS-DiD on total operating revenue and non-Medicaid inpatient days. Neither should move differentially with Medicaid SDP receipt if the causal pathway is Medicaid-specific.

2. Random-dose permutation: in 200 draws we randomly reassign SDP treatment across HCRIS states (preserving the 41/13 treatment-ratio), re-estimate the ATT, and report the 2.5/97.5 percentile of the null distribution.

3. Net-of-tax sensitivity: at each of five recapture rates {0.10, 0.16, 0.23, 0.29, 0.35}, we regress the state-level post-minus-pre first difference in the outcome on per-state net-of-tax dose (in $B) and report the implied ATT ± 95% CI. This directly surfaces how much the headline shifts as the state-prior recapture rate moves across its plausible span.

4. Alternative specifications: (a) alternative winsorization (5th/95th); (b) alternative bandwidth (2018–2025 and 2014–2025); (c) never-treated controls only (drop not-yet-treated); (d) inpatient-hospital-specific cohort as the treatment key.

4.5 Heterogeneity by provider class

For each of the four in-scope SDP provider classes (inpatient hospital, outpatient hospital, nursing facility, AMC professional), we rebuild the state-level g using only cohorts in that class and re-run the CS-DiD ATT. This separates whether the headline effect is concentrated in the classes that HCRIS best captures (inpatient hospital) vs. channels where HCRIS has weaker signal (AMC professional).

4.6 §71116 descriptive supplement

As documented in §3.5, the §71116 dose has no identifying variation in the public corpus. We therefore frame the §71116 analysis descriptively rather than causally, reporting three facts: (i) 9 parseable arrangements with dose = 0; (ii) 21 of 26 arrangements CMS-designated as likely grandfathered under §71116(b); (iii) 4 arrangement-specific net-of-tax cases. We interpret this as a *placebo confirmation that the §71116 cap is not yet binding* in the public record — not as a causal result.

4.7 Inference

All CS-DiD standard errors use analytic influence-function approximations on the 2×2 within-cell first differences. TWFE / Sun-Abraham specifications use state-clustered standard errors. The random-dose permutation null provides a nonparametric benchmark that does not rely on within-hospital clustering assumptions.

4.8 Implementation

All analysis is implemented in Python 3.11 with `pandas`, `numpy`, `statsmodels`, `pyfixest`, and `matplotlib`. The entire Phase 4 analysis is reproducible from the clean data layer via a single shell script (`analysis/run_all.sh`), with header blocks and module-level docstrings documenting every function. No R is required; no Stata `.do` files are used.

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5. Results

5.1 CS-DiD headline: operating margin

Table 2 reports the headline CS-DiD overall ATT on the four primary outcomes. On the winsorized operating margin, the pooled ATT is −0.006 (SE 0.008, 95% CI −0.022, +0.011) under the binary-treatment specification. Weighting by gross state-level TC yields ATT = −0.004 (SE 0.008). Weighting by net-of-tax TC — our preferred specification per the Baicker-Staiger (2005) lesson — yields ATT = −0.003 (SE 0.008, 95% CI −0.019, +0.013). In all three specifications, the CS-DiD ATT on operating margin is statistically indistinguishable from zero and the 95% CI rules out an effect larger than ±2.2 percentage points in either direction.

Figure 2a shows the event-study dynamic ATT(e) for e □ [−9, +2]. Pre-period leads (e = −9 through e = −1) are individually imprecise but the magnitude pattern does not reveal a clear treatment-leading trend: the largest pre-period point estimate is −0.029 at e = −6 with a 95% CI of (−0.059, −0.0000). Post-treatment lags at e = 0, 1, 2 are −0.008 (SE 0.012), +0.004 (SE 0.013), and −0.023 (SE 0.023), respectively. The post-period pattern is visually flat around zero.

The random-dose permutation null (Figure 6, 200 draws) gives a 2.5/97.5 percentile envelope of (−0.013, +0.008). The observed binary-treatment ATT of −0.006 sits comfortably inside this null, confirming that the headline null is not an artifact of a degenerate placebo distribution. The Rambachan-Roth HonestDiD robust confidence sets at M = 1.0 (Table A2, Figure 4) are (−0.061, +0.044) at e = 0, (−0.052, +0.060) at e = 1, and (−0.097, +0.052) at e = 2 — comfortably straddling zero in every event-time. The M = 0.25 bound is tightest — (−0.039, +0.022) at e

= 0 — and still straddles zero. We conclude that the operating-margin null is robust to plausible parallel-trends deviations.

5.2 Divergence with Borusyak-Jaravel-Spiess and the TWFE benchmark

The Borusyak-Jaravel-Spiess imputation estimator (Table A3) returns an overall ATT of +0.021 (SE 0.002, 95% CI +0.016, +0.026) on operating margin, markedly larger and of opposite sign to the CS-DiD estimate. The static TWFE benchmark is +0.017 (SE 0.009, 95% CI −0.002, +0.035), between CS-DiD and BJS.

Two features of the research design help interpret this divergence. First, the cohort structure is degenerate: 41 of 43 treated states have $g = 2023$ and only 2 have $g = 2024$ or later. BJS fits unit + year fixed effects on the *untreated* sub-sample (13 never-treated states) and then extrapolates Y(0) forward for the treated observations. When the untreated sub-sample is small relative to the treated sub-sample and the post-period is short, BJS can extrapolate aggressively along secular trends, producing a positive ATT if aggregate operating margin would have fallen in the counterfactual (which it plausibly would have, given post-COVID margin compression). CS-DiD, by contrast, computes 2×2 comparisons within cohort and is robust to the treated/untreated imbalance at the cost of efficiency.

Second, the Goodman-Bacon decomposition (Table A4, Figure 5) confirms that 84% of identifying weight in the TWFE estimator comes from the "2023 vs. never" clean 2×2 comparison, with only 7% from contaminated (already-treated-as-control) comparisons. This is reassuring in the sense that TWFE contamination is small; it also confirms that any divergence between BJS and CS-DiD cannot be attributed to TWFE negative-weight artifacts.

We lead with CS-DiD because (a) its 2×2 comparison weights are transparent, (b) HonestDiD bounds on CS-DiD straddle the BJS point estimate at M ⍰ 1.0, and (c) the permutation-null envelope places the CS-DiD estimate inside the null while the BJS estimate is outside it — but BJS extrapolates in exactly the way that the permutation null does not. The honest conclusion is that we cannot distinguish a true-null operating-margin effect from a modest positive effect swamped by post-COVID noise; this is a central caveat we revisit in §6.

5.3 The AMC-professional heterogeneity signal

The most credible signal in the paper is the provider-class heterogeneity result (Table 3, Figure 8a). Rebuilding the state-level g using only arrangements in each of the four in-scope classes yields:

- AMC-professional: ATT = +0.010 (SE 0.005, 95% CI +0.001, +0.020), n = 22 states contributing a post-Feb-2023 AMC-professional arrangement.
- Inpatient hospital: ATT = +0.007 (SE 0.005, 95% CI −0.003, +0.017), n = 38 states.
- Nursing facility: ATT = +0.002 (SE 0.004, 95% CI −0.007, +0.010), n = 17 states.
- Outpatient hospital: ATT = −0.002 (SE 0.006, 95% CI −0.013, +0.009), n = 12 states.

Only the AMC-professional estimate has a lower-CI bound above zero, and it is the only class with a positive point estimate and a 95% CI excluding zero. The inpatient-hospital estimate is positive and near-significant but imprecise. The nursing-facility and outpatient-hospital estimates are null.

This pattern is consistent with the institutional structure of the AMC-professional SDP channel. AMC-professional arrangements flow to academically-affiliated teaching hospitals through "Professional Services at an Academic Medical Center" preprints that pay uplift to the faculty-practice-plan professional-fee schedule. These payments typically pass through an MCO-to-AMC-to-faculty-practice contract chain with relatively small recapture through provider taxes (faculty practice plans are not typically subject to the state hospital-tax base). Net-of-tax bite in AMC-professional arrangements is therefore closer to the gross TC than in inpatient hospital arrangements, where 24.7% average state-prior recapture applies. The +1-percentage-point ATT on operating margin at AMC-affiliated hospitals is the paper's clearest causal signal, and we disclose it with appropriate caveats: 22-state n, 3-year post-period, and the general-volume-channel concern raised by the placebo-outcome analysis below.

5.4 Placebo outcomes: the honest caveat

The placebo-outcome analysis delivers the paper's central methodological caveat. Total operating revenue — which is not Medicaid-specific and should not respond to a Medicaid-specific transfer — has an estimated ATT of +\$16.7M per hospital-year (SE \$5.4M, 95% CI +\$6.0M, +\$27.3M). Non-Medicaid inpatient days — constructed as `total_ip_days × (1 - medicaid_share_days)` — has an estimated ATT of +545 days per hospital-year (SE 205, 95% CI +144, +947).

Neither placebo is null. The pattern is consistent with a general-volume channel: hospitals in states that approved SDPs after February 2023 appear to have experienced modestly higher overall patient volume and revenue in the post-period than hospitals in the 13 never-treated states. This could reflect (a) differential post-COVID recovery trajectories correlated with state-level Medicaid policy vigor, (b) selection on unobservables where states approving SDPs were also states with stronger underlying hospital demand, or (c) a true SDP-driven expansion of hospital capacity that served both Medicaid and non-Medicaid patients.

We explicitly do not attempt to discriminate among these explanations with the current data. The disciplinary implication is that we cannot claim a Medicaid-specific margin effect — any margin channel identified in the main specification is equally plausibly explained by the same general-volume mechanism that the placebo outcomes capture. This is the honest reading of the full result package.

We also note that the total-inpatient-days ATT of +508 days (Table 2) is barely larger than the non-Medicaid-days ATT of +545 days, which mechanically implies that Medicaid days did not rise (or rose only modestly). Correspondingly, the Medicaid share of inpatient days is null: ATT = −0.00003 (SE 0.002) under binary treatment and +0.001 (SE 0.002) under net-weighted treatment. The CS-DiD design is not detecting a Medicaid-specific shift in hospital activity.

5.5 Net-of-tax sensitivity

Table 4 and Figure 7 report the net-of-tax sensitivity analysis. At recapture-rate priors of {0.10, 0.16, 0.23, 0.29, 0.35}, the implied dose-response slope on operating margin rises mechanically from 0.0006 per \$B net-of-tax dose (recapture 10%) to 0.0008 per \$B (recapture 35%), with state-level 95% CIs that include zero throughout. The implied overall ATT (state-level dose-weighted first-difference regression) is +0.003 (95% CI −0.005, +0.011) across the full span. The headline is invariant to the net-of-tax state prior within the plausible range.

This is a consequential robustness result. It rules out a naive Baicker-Staiger "phantom-transfer" story in which the pooled null masks a genuinely large effect attenuated by recapture: the span of priors we consider plausible (10% to 35%) simply does not move the headline enough to cross the zero threshold. The null survives the Baicker-Staiger robustness. What it does not survive is the observation that arrangement-specific recapture data are available for only 2% of dollar exposure; a successful CMS FOIA response would allow a tighter test.

5.6 Alternative specifications

Table A5 reports the alternative-specification battery. The main CS-DiD ATT on operating margin is stable across (a) 5th/95th winsorization (−0.005, SE 0.008), (b) alternative bandwidth 2018–2025 (−0.007, SE 0.009), (c) never-treated-only controls (−0.008, SE 0.010), and (d) inpatient-hospital-only cohort assignment (+0.007, SE 0.005, bordering significance as the direction flips — this is the same signal as the inpatient-hospital class in Table 3). No specification moves the headline outside (−0.015, +0.010).

5.7 §71116 descriptive supplement

Figure 9 visualizes the §71116 hand-coding results. Of the 26 in-scope arrangements:

- 9 of 26 have parseable Table 2.A data. Dose = 0 in all 9. The maximum parsed pre-cap rate is 101.3% of Medicare (non-expansion state, cap = 110%). No arrangement has a positive binding dose.
- 21 of 26 are CMS-designated as likely grandfathered under §71116(b). These arrangements are frozen at current payment levels until the 2028 phase-down begins.

- 17 of 26 have no populated Table 2.A: 14 lack the heading entirely and 3 have the heading but blank row fields.

For the 4 arrangement-specific net-of-tax cases (3 LA inpatient-and-outpatient and 1 FL inpatient-and-outpatient), the parsed Addendum Table 4.A IGT recapture estimates are 18%, 24%, 28%, and 32% respectively — consistent with, but not identical to, our hand-coded state-level priors for Louisiana (0.22) and Florida (0.25). This narrow comparison offers no basis for generalization but is consistent with the state-level-prior approach not being systematically biased.

The §71116 event-study estimate is therefore not identified in currently-available data. We interpret this as a placebo confirmation that the cap is not yet binding in the public record, and we defer causal estimation of §71116 to a follow-on paper once the FOIA response is received and HCRIS data for fiscal year 2026+ are available.

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6. Discussion

6.1 Interpretation of the pooled null

The central empirical finding of this paper is a cautiously null pooled ATT on hospital operating margin. The null is robust to the choice of CS-DiD versus Sun-Abraham IW, to net-of-tax weighting, to HonestDiD bounds at $M = 1.0$, to 5th/95th winsorization, to alternative bandwidth, to dropping not-yet-treated controls, and to the random-dose permutation benchmark. The null is *not* robust to the BJS imputation estimator, which returns +0.021 (SE 0.002) and reflects forward-extrapolation along the never-treated-state trend. Given the degenerate single-cohort structure and the short post-period, we interpret the BJS result as an efficiency-bias tradeoff rather than a contradiction of the CS-DiD null.

The null has three non-mutually-exclusive economic interpretations. First, SDPs may in fact have small short-run financial effects on the typical acute-care hospital because the transfer is partially recaptured through provider taxes and IGTs (the Baicker-Staiger 2005 channel). Our net-of-tax sensitivity over 10%–35% recapture priors does not rule this out, though the fact that the headline is stable across the full span suggests the Baicker-Staiger attenuation is not the decisive mechanism [@baicker2005fiscal]. Second, the post-period is short (2–3 years depending on cohort) and the single-cohort structure is thin; a longer and deeper pre/post panel (unlocked by a successful FOIA response) would narrow the confidence interval and might reveal a small positive effect. Third, the Medicaid-share-of-volume null and the positive placebo outcomes are consistent with a general-volume channel where SDPs contribute to capacity expansion that serves both Medicaid and non-Medicaid patients, but do not differentially improve the Medicaid bottom line. We think the third channel is the most plausible reading, but we do not have the identification to distinguish it from the first two.

6.2 Relation to the Medicaid-financing literature

Our pooled null stands in interesting tension with the ACA-Medicaid-expansion hospital-finance literature. Dranove, Garthwaite, and Ody (2016) and Lindrooth, Perraillon, Hardy, and Tung (2018) found substantial improvements in operating margins and reduced closure risk in expansion states, concentrated at high-uncompensated-care hospitals [@dranove2016uncompensated; @lindrooth2018hospital]. Finkelstein, Hendren, and Luttmer (2019) estimated that roughly 60% of the marginal Medicaid dollar flows to providers as uncompensated-care offset [@finkelstein2019value]. These results establish that meaningful Medicaid revenue changes do move hospital financial outcomes measurable in HCRIS. Our null on SDPs despite their fiscal scale suggests the SDP mechanism differs structurally from ACA-expansion-style coverage gains. The most plausible explanations, consistent with Baicker and Staiger (2005), are (a) larger net-of-tax recapture than gross numbers imply and (b) less targeting on high-uncompensated-care hospitals than ACA-expansion coverage gains mechanically delivered [@baicker2005fiscal].

On the other hand, our AMC-professional heterogeneity result echoes the physician-fee literature. Alexander and Schnell (2024) show that exogenous Medicaid fee increases raise physician office-visit availability and close roughly half of the Medicaid-private access gap; Clemens and Gottlieb (2017) document that public payment changes transmit to private-sector pricing [@alexander2024impacts; @clemens2017shadow]. Both papers support the interpretation that provider-level financial targeting — particularly when the provider type is professional services rather than hospital services — does translate into real financial outcomes. Hackmann (2019)'s parallel result for nursing homes (a 10% Medicaid rate increase raised skilled-nurse staffing by 8.7%) would suggest we should see a similar effect at the nursing-facility class [@hackmann2019incentivizing]; we do not. One explanation is that nursing-facility SDPs in the public corpus are dominated by hold-harmless provider-tax arrangements, which recapture a large share of the gross transfer. A second is simply statistical power: n = 17 states for the nursing-facility heterogeneity run is thin.

Duggan (2004) provides the relevant cautionary precedent. His California county-level mandate design showed that contracting out to managed care produced no detectable health improvement for beneficiaries but raised Medicaid spending by ~12% [@duggan2004contracting]. If SDPs are channeling nominal capitation dollars to hospital balance sheets without a corresponding change in real hospital activity or Medicaid beneficiary outcomes, Duggan's null result is the relevant benchmark. Our Medicaid-share-of-volume null is consistent with this reading. We note, however, that pure "capitation arithmetic" interpretations of SDPs run into the fact that SDPs have specific provider-class targeting and rate-methodology documentation that materially restricts MCOs' discretion; they are not a pure capitation loop.

6.3 The placebo-outcome channel and the limits of the SDP claim

The placebo-outcome pattern — positive ATTs on total operating revenue and non-Medicaid inpatient days — deserves separate discussion. In a conventional DiD evaluation of a Medicaid-specific policy, non-null placebo outcomes would be grounds to reject the design entirely. We take a different view here because we believe the correct interpretation is not a rejection of the CS-DiD identification but rather a disciplined narrowing of the claim. SDPs are not a Medicaid-specific policy shock in the same sense that, say, a Medicaid-fee-schedule increase is. SDPs flow through managed-care capitation pools, and any hospital-level financial response is mediated by the MCO's rate-setting, the state's provider-tax structure, and the hospital's broader patient mix. It is entirely plausible that SDP approvals correlate with other contemporaneous state Medicaid-policy activity (fee-schedule updates, 1115 waivers, health-related social-needs authorities) that collectively raise the state's Medicaid-policy vigor and, by extension, the revenue environment for hospitals serving large Medicaid populations.

The correct claim is therefore the *narrower* claim that we observe no Medicaid-specific margin effect on the typical acute-care hospital. The *broader* claim — that SDPs causally expand general hospital activity — is one the placebo pattern is suggestive of but that the current identification cannot credibly isolate. A follow-on design that explicitly instruments for state-SDP vigor (e.g., using pre-2016 UPL exposure as an instrument, exploiting the 2016 rule as the forcing variable) would be necessary to separate SDP-specific causation from general Medicaid-policy-vigor correlation.

6.4 Policy implications for OBBBA §71116

The policy relevance of this paper's null is somewhat counterintuitive. OBBBA §71116 was enacted on the assumption — documented in CBO and congressional-staff scoring — that pre-OBBBA SDPs were producing large implicit federal subsidies to states and, by extension, to hospital balance sheets. Our pre-OBBBA null on operating margin does not refute that scoring: federal SDP expenditures are real ($110 billion/year per MACPAC), and those dollars reach hospitals in some form. What our results suggest is that (a) a large fraction of the gross flow is recaptured through state provider-tax arrangements before it lands in hospital operating margins (the Baicker-Staiger mechanism), (b) the residual net transfer appears to support general hospital capacity rather than a Medicaid-specific subsidy, and (c) the specific provider-class channel (inpatient hospital vs. AMC professional vs. nursing facility) matters materially for how the net transfer expresses in financial outcomes.

For the §71116 implementation, four specific implications follow:

1. The grandfathering provision is doing serious work. 21 of 26 in-scope arrangements in the public corpus are CMS-designated as likely grandfathered under §71116(b). The immediate fiscal effect of the cap on hospital margins is therefore muted through 2028. Any CBO-style scoring of the short-run margin impact should reflect this.

2. The cap is unlikely to produce dramatic inpatient-hospital margin reductions on average. Our pooled pre-OBBBA null implies that the reverse direction — removing the pre-OBBBA SDP — is also unlikely to produce dramatic margin reductions. The extrapolation is not tight (pre-OBBBA data are left-censored in the public record) but it is the most defensible base-case prediction.
3. AMC-professional channels are the most at-risk. The provider class that shows the clearest positive pre-OBBBA effect in our data is precisely the class that is most exposed to §71116's professional-fee cap, because AMC-professional SDPs typically benchmark to a percent of Medicare Part B fee-schedule rates and are often above the 100% cap. A meaningful share of the grandfathered AMC-professional arrangements will lose margin in 2028+.
4. States with heavy ACR benchmarking face the steepest cliff. Approximately 10% of our panel references ACR benchmarking, which Marr et al. (2023) document as ▯185% of Medicare on average [@marr2023hospital]. These are the arrangements where §71116's Medicare-referenced cap bites hardest, and they are concentrated in a small number of states. A state-by-state §71116-exposure map, combining our net-of-tax data with the Marr et al. ACR-to-Medicare conversion, is a natural follow-on.

6.5 Limitations

We catalogue seven central limitations, drawing on the Agent 4B strengths-and-limitations memo (`analysis/robustness/streng`

6.5.1 Primary-to-secondary identification pivot. The pre-specified primary design was the §71116 event study. Hand-coding of 26 in-scope arrangements revealed no binding dose variation in the public corpus (§3.5). We pivoted to CS-DiD on first-SDP-approval timing as the primary design. The user authorized this pivot on 2026-04-16. The §71116 analysis is retained as a descriptive supplement.

6.5.2 Net-of-tax state-prior limitation. Only 4 of 1,038 arrangements (0.4% by count, 2.0% by dollar exposure) have arrangement-specific Addendum Table 4.A IGT data. The remaining 99.5% relies on a state-level MACPAC/KFF recapture prior. The §5.5 sensitivity spans the full plausible prior range but does not resolve the fundamental measurement gap. Every sentence in this paper that references *magnitude* of the treatment is qualified by this limitation [@baicker2005fiscal].

6.5.3 Pending CMS FOIA. A FOIA letter (`data/foia-request-letter-cms.md`) requests the unredacted Excel Addenda workbooks (Tables 1.A–8.A) for all approved SDP preprints 2017–2026. A successful FOIA response would unblock (a) the §71116 event study with true binding-cap variation, (b) arrangement-specific net-of-tax coding for all 1,038 arrangements, and (c) pre-2023 first-approval history that would resolve the cohort-timing censoring problem. Expected turnaround: 6–12 months.

6.5.4 Thin timing variation. The current CS-DiD is effectively a single-cohort (2023) DiD with 13 never-treated states. HonestDiD bounds are therefore critical for pre-trends interpretation, and we rely on them rather than on point-level parallel-trends tests.

6.5.5 Imprecise pre-trends. With 13 never-treated controls, some pre-period leads ($e = -9$ has 1 contributing cohort) have wide CIs. The point-estimate magnitudes of pre-period leads are within ± 0.03 on operating margin but we do not rely on formal parallel-trends tests — HonestDiD is the appropriate tool.

6.5.6 Hospital-provider-class mismatch. HCRIS is at the hospital-year level; our SDP cohorts are at the state $\times$ provider-class level. The main specification assigns each hospital the earliest SDP cohort in its state across all provider classes. The inpatient-hospital-only robustness run yields ATT = +0.007 (SE 0.005), consistent in direction with the Table 3 class-specific result.

6.5.7 BJS-CS-DiD divergence. BJS imputation yields ATT = +0.021 (SE 0.002), of opposite sign to CS-DiD. We attribute this to extrapolation bias given the degenerate cohort structure and the short post-period. We lead with CS-DiD because its 2×2 comparison weights are transparent, its HonestDiD bounds are well-defined, and the permutation null is consistent with the CS-DiD estimate. We report BJS in the robustness appendix for full disclosure.

6.6 Future research

The most important follow-on is the CMS FOIA-contingent §71116 event study targeting Q2 2027 field-date. Three other extensions are attractive. First, a state-by-state §71116-exposure map combining our net-of-tax data with the Marr et al. (2023) ACR-to-Medicare conversion, to identify the states most at risk from the 2028 phase-down. Second, an AMC-professional-specific analysis using the AAMC hospital identifier and faculty-practice-plan revenue data, to isolate the channel that drives the pooled positive heterogeneity signal we document in §5.3. Third, a hospital-closure outcome analysis using AHA Annual Survey service-line data (purchase gate pending at ~$10K), to test whether SDP exposure reduces hospital-closure probability at high-uncompensated-care hospitals, analogous to Lindrooth et al. (2018) [@lindrooth2018hospital].

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7. Conclusion

This paper provides the first quasi-experimental evaluation of state-directed payments in Medicaid managed care — the largest unevaluated spending category in the Medicaid program. Using a novel panel of every CMS-approved SDP preprint from February 2023 onward, a Callaway-Sant'Anna staggered DiD on first-approval cohorts, and a full robustness battery including Sun-Abraham IW, Borusyak-Jaravel-Spiess imputation, Goodman-Bacon decomposition, Rambachan-Roth HonestDiD bounds, random-dose permutation inference, and net-of-tax sensitivity, we estimate a pooled ATT on hospital operating margin of −0.006 percentage points (SE 0.008), statistically indistinguishable from zero. The signal that survives is a positive heterogeneity effect on academic-medical-center-professional arrangements (ATT = +0.010, 95% CI +0.001, +0.020) and a positive but imprecise effect on inpatient hospital services (+0.007). The caveat that disciplines the claim is a non-null placebo pattern on total operating revenue and non-Medicaid inpatient days, consistent with a general-volume channel rather than a Medicaid-specific cost-shifting mechanism.

The §71116 event study — the originally pre-specified primary design — is not currently fieldable because the public CMS corpus contains no arrangement with a binding §71116 dose: 9 of 26 in-scope arrangements parse to dose = 0, 21 of 26 are CMS-designated as likely grandfathered under §71116(b), and 17 of 26 have no populated Addendum Table 2.A because the Excel workbooks are not uniformly merged into approval-package PDFs. A CMS FOIA request for the unredacted workbooks is pending. Combined with HCRIS data for fiscal year 2026, a successful FOIA response would make a §71116 event study fieldable around Q2 2027.

The paper's core contributions are (1) the first longitudinal arrangement-level SDP panel with a net-of-tax treatment variable, (2) the first quasi-experimental null on the pre-OBBBA regime with appropriate HonestDiD and permutation discipline, (3) documentation of the Baicker-Staiger "phantom transfer" limitation as a first-order measurement problem in the public CMS corpus, and (4) the methodological template for the §71116 event study that will become the natural follow-on paper. For policy, our pre-OBBBA null suggests that the §71116 cap is unlikely to produce dramatic inpatient-hospital margin reductions on average, with the caveat that AMC-professional and ACR-benchmarked arrangements are the most at-risk channels. The largest uncertainty is structural: the CMS public corpus is designed for regulatory transparency at the arrangement level, not for empirical analysis. Unlocking the Excel Addenda workbooks is the most valuable single piece of follow-on data work.

Tables

Table 1. Summary statistics of the CMS-approved SDP preprint panel

Category	Value
Total unique SDP preprints	1,038
Unique states plus territories	43 + Puerto Rico
Unique CMS-labeled provider classes	11
Rating-period window	2020Q4 – 2026Q2
Gross projected total computable (sum)	$296.2 B

Category	Value
Net-of-provider-tax total computable (sum)	$223.0 B
Implied average recapture rate	24.7%
Fee-schedule preprints	894
Value-based preprints	144
Renewals / New / Amendments	737 / 172 / 129
§71116 in-scope arrangements (4 classes)	547
§71116 active-treatment (non-grandfathered, post-Jul-2025)	26
Arrangements with arrangement-specific Table 4.A IGT data	4
Arrangements with arrangement-specific Table 5.A provider-tax data	0

Source: `data/clean/sdp_panel.parquet`, `data/clean/sdp_net_of_tax.csv`, `data/summary-statistics.md`.

Table 2. Callaway–Sant'Anna overall ATT on hospital outcomes

Outcome	Dose	ATT	SE	95% CI	n cells
Operating margin (win.)	Binary	−0.0056	0.0083	(−0.022, +0.011)	6
Operating margin (win.)	Gross-weighted	−0.0041	0.0083	(−0.020, +0.012)	6
Operating margin (win.)	Net-weighted	−0.0030	0.0083	(−0.019, +0.013)	6
Uncomp-care ratio	Binary	+0.00006	0.00058	(−0.001, +0.001)	1
Uncomp-care ratio	Net-weighted	−0.00004	0.00054	(−0.001, +0.001)	1
Total IP days	Binary	+508	209	(+99, +918)	6
Total IP days	Net-weighted	+699	212	(+283, +1,115)	6
Medicaid share of IP days	Binary	−0.00003	0.0020	(−0.004, +0.004)	6
Medicaid share of IP days	Net-weighted	+0.0014	0.0020	(−0.003, +0.005)	6

Source: `analysis/tables/csdid_overall_summary.csv`. *ATT estimates aggregated over all (g, t) with e ⍰ 0.*

Table 3. Provider-class heterogeneity: CS-DiD ATT on operating margin (winsorized)

Provider class	ATT	SE	95% CI	n states
Inpatient hospital	+0.0072	0.0050	(−0.003, +0.017)	38
Outpatient hospital	−0.0023	0.0056	(−0.013, +0.009)	12
Nursing facility	+0.0019	0.0044	(−0.007, +0.010)	17
AMC-professional	+0.0103	0.0048	(+0.001, +0.020)	22

Source: `analysis/tables/csdid_heterogeneity_operating_margin_win.csv`. *State-level g rebuilt from class-specific cohort.*

Table 4. Net-of-tax sensitivity: implied slope and ATT on operating margin

Recapture rate	Slope per $B	Slope SE	Overall ATT	95% CI
0.10	0.00058	0.00083	+0.0029	(−0.005, +0.011)
0.16	0.00063	0.00089	+0.0029	(−0.005, +0.011)
0.22	0.00068	0.00096	+0.0029	(−0.005, +0.011)
0.29	0.00074	0.00105	+0.0029	(−0.005, +0.011)
0.35	0.00081	0.00115	+0.0029	(−0.005, +0.011)

Source: analysis/robustness/nettax_sensitivity_operating_margin_win.csv. State-level first-difference regression of Δoperating-margin on net-of-tax $B.

Table 5. Placebo outcomes: CS-DiD overall ATT

Outcome	ATT	SE	95% CI
Total operating revenue ($)	+$16.66 M	$5.43 M	(+$6.01 M, +$27.31 M)
Non-Medicaid IP days	+545	205	(+144, +947)

Source: analysis/robustness/placebo_outcomes.csv. Both placebo outcomes are not null, consistent with a general-volume channel rather than Medicaid-specific cost-shifting.

Table 6. Robustness summary: alternative estimators on operating margin (winsorized)

Estimator	ATT	SE	95% CI
CS-DiD binary (primary)	−0.0056	0.0083	(−0.022, +0.011)
CS-DiD net-weighted	−0.0030	0.0083	(−0.019, +0.013)
Sun-Abraham IW / TWFE static	+0.0166	0.0094	(−0.002, +0.035)
Borusyak-Jaravel-Spiess	+0.0209	0.0024	(+0.016, +0.026)
Permutation null 95% envelope	—	—	(−0.0126, +0.0082)
HonestDiD M = 1.0 robust CI at e = 1	—	—	(−0.052, +0.060)

Source: analysis/robustness/robustness_summary.csv, analysis/robustness/honestdid_operating_margin_win.csv.

Figures

Figure 1. Raw cohort-specific outcome trends, 2011–2025, for the four main outcomes. Treated states' cohort-mean hospital-year trajectories are plotted against the never-treated control mean, with a vertical treatment-onset marker at 2022.5. File: analysis/figures/fig01_raw_trends_<outcome>.png (4 panels: operating margin, uncompensated-care ratio, total IP days, Medicaid IP share).

Figure 2. CS-DiD event study: dynamic ATT(e) with 95% pointwise CI bands for event times e ▯ [−9, +2], for each of the four main outcomes under binary treatment. Vertical line at e = 0 marks treatment onset. File: analysis/figures/fig02_eventstudy_<outcome>_binary.png (4 panels).

Figure 3. Treatment timing distribution: state-level first-approval year (41 states at 2023, 1 at 2024, 1 at 2025) and state × provider-class first-approval quarter (205 cohorts, 2023Q1 through 2026Q1). File: analysis/figures/fig03_cohort_support.png.

Figure 4. Rambachan-Roth HonestDiD relative-magnitudes robust confidence sets on operating margin for M ⍰ {0.25, 0.5, 1.0, 1.5, 2.0} at event times e ⍰ {0, 1, 2}. File: `analysis/figures/fig04_honestdid_operating_margin_win.png`.

Figure 5. Goodman-Bacon decomposition scatter on operating margin: 2×2 comparison estimates plotted against weight share, color-coded clean (never-treated as control) versus contaminated (already-treated as control). 84% of identifying weight is from the "2023 vs. never" clean comparison. File: `analysis/figures/fig05_bacon_operating_margin_win.png`.

Figure 6. Placebo outcomes panel (total operating revenue, non-Medicaid IP days) with 200-draw random-dose permutation null distribution overlaid on the observed ATT. File: `analysis/figures/fig06_placebo_panel_operating_margi`

Figure 7. Net-of-tax sensitivity forest plot: per-$B net-of-tax dose slope on operating margin across recapture-rate priors {0.10, 0.16, 0.22, 0.29, 0.35}. File: `analysis/figures/fig07_nettax_forest_operating_margin_win.png`.

Figure 8. Provider-class heterogeneity forest: CS-DiD ATT on each outcome by provider class (inpatient hospital, outpatient hospital, nursing facility, AMC professional). File: `analysis/figures/fig08_heterogeneity_<outcome>.png` (4 panels).

Figure 9. §71116 hand-coding descriptive supplement: (a) dose distribution over 9 parseable arrangements (all dose = 0); (b) grandfathered-vs-non-grandfathered split (21 vs. 5); (c) parseable-vs-unparseable split (9 vs. 17); (d) arrangement-specific net-of-tax coverage (4 of 200 top by gross TC). File: `analysis/figures/fig09_s71116_descriptive.png`.

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Appendix — State-Directed Payments in Medicaid Managed Care

Paper: `state-directed-payments-panel/` Companion to: `manuscript/draft.md` Date: 2026-04-16

This appendix reports the full Callaway-Sant'Anna cohort grids, HonestDiD sensitivity tables, Sun-Abraham and Borusyak-Jaravel-Spiess comparison tables, Goodman-Bacon decomposition weights, §71116 hand-coding notes, and alternative-specification tables referenced in the main text.

A1. CS-DiD cohort support and state-level treatment

A1.1 State × provider-class cohorts by first-approval quarter

First-approval quarter	N state × class cohorts
2023Q1	42
2023Q2	38
2023Q3	38
2023Q4	28
2024Q1	13
2024Q2	6
2024Q3	5
2024Q4	10
2025Q1	2
2025Q2	8

First-approval quarter	N state × class cohorts
2025Q3	6
2025Q4	3
2026Q1	6

Source: `data/clean/sdp_csdid_treatment.csv`*. Total: 205 state × provider-class cohorts.*

A1.2 State-level treatment summary

Of the 43 SDP states in the panel, 41 have earliest approval year g = 2023, 1 has g = 2024 (District of Columbia), and 1 has g = 2025 (Colorado). The 13 HCRIS states without a CMS-public SDP approval in the panel window (AK, AL, AS, CT, GU, ID, ME, MP, MT, ND, SD, VI, WY) serve as the never-treated comparison group.

Source: `analysis/tables/state_treatment_cohorts.csv`*.*

A2. CS-DiD event-study dynamic ATT: full cohort grids

A2.1 Operating margin (winsorized), binary treatment

e	ATT	SE	n cells	95% CI lo	95% CI hi
−9	+0.0280	0.0468	1	−0.0637	+0.1197
−8	+0.0116	0.0427	2	−0.0721	+0.0952
−7	−0.0190	0.0153	3	−0.0490	+0.0109
−6	−0.0295	0.0150	3	−0.0589	−0.0000
−5	−0.0091	0.0154	3	−0.0394	+0.0211
−4	−0.0236	0.0137	3	−0.0504	+0.0032
−3	−0.0041	0.0162	3	−0.0358	+0.0275
−2	−0.0198	0.0116	3	−0.0426	+0.0030
0	−0.0084	0.0117	3	−0.0312	+0.0145
1	+0.0038	0.0135	2	−0.0226	+0.0302
2	−0.0227	0.0229	1	−0.0675	+0.0221

Source: `analysis/tables/csdid_eventstudy_operating_margin_win_binary.csv`*. Reference period e = −1 is omitted (reference category). Pre-period e = −9 has only 1 contributing cohort and wide CI; this is the structural limitation documented in §6.5.5 of the main text.*

A2.2 Other outcomes

Full event-study grids for uncompensated-care ratio, total inpatient days, and Medicaid share of inpatient days are in: - `analysis/tables/csdid_eventstudy_uncomp_care_ratio_binary.csv` - `analysis/tables/csdid_eventstudy_total` - `analysis/tables/csdid_eventstudy_medicaid_share_days_binary.csv`

Each grid reports ATT(e), SE, n cells, and 95% CI for e ⍰ {−9, …, +2}, with parallel gross-weighted and net-weighted variants in the `*_gross_weighted.csv` and `*_net_weighted.csv` files.

A3. Rambachan-Roth HonestDiD bounds: full table

Robust 95% confidence sets under relative-magnitudes parallel-trends deviation M ⍰ {0.25, 0.5, 1.0, 1.5, 2.0} × largest absolute pre-period coefficient (|ATT(−6)| = 0.0295). Outcome: operating margin (winsorized).

M	e	ATT	Robust CI lo	Robust CI hi
0.25	0	−0.0084	−0.0386	+0.0219
0.25	1	+0.0038	−0.0300	+0.0376
0.25	2	−0.0227	−0.0749	+0.0295
0.50	0	−0.0084	−0.0460	+0.0292
0.50	1	+0.0038	−0.0373	+0.0449
0.50	2	−0.0227	−0.0823	+0.0368
1.00	0	−0.0084	−0.0607	+0.0439
1.00	1	+0.0038	−0.0521	+0.0597
1.00	2	−0.0227	−0.0970	+0.0516
1.50	0	−0.0084	−0.0754	+0.0587
1.50	1	+0.0038	−0.0668	+0.0744
1.50	2	−0.0227	−0.1117	+0.0663
2.00	0	−0.0084	−0.0901	+0.0734
2.00	1	+0.0038	−0.0815	+0.0891
2.00	2	−0.0227	−0.1265	+0.0810

Source: `analysis/robustness/honestdid_operating_margin_win.csv`. At every M and every event time, the robust 95% CI straddles zero, confirming that the operating-margin null is robust to plausible parallel-trends deviations up to 2× the largest pre-period coefficient.

A4. Sun-Abraham and Borusyak-Jaravel-Spiess comparison

A4.1 Static TWFE and BJS overall ATT

Estimator	ATT	SE	95% CI	N
TWFE static (benchmark)	+0.01655	0.00943	(−0.002, +0.035)	42,804
BJS imputation	+0.02094	0.00235	(+0.016, +0.026)	6,750

Source: `analysis/robustness/twfe_static_operating_margin_win.csv`, `analysis/robustness/bjs_overall_operatin`

A4.2 Sun-Abraham IW event study

See `analysis/robustness/twfe_eventstudy_operating_margin_win.csv` for event-time-specific Sun-Abraham IW estimates with state-clustered standard errors. The IW event-study profile is qualitatively similar to the CS-DiD profile shown in Table A2.1, with slightly wider CIs due to the cluster-robust variance estimator.

A4.3 BJS event study

See `analysis/robustness/bjs_eventstudy_operating_margin_win.csv` for BJS imputation-based event-time coefficients. BJS returns a positive and statistically distinguishable profile from $e = 0$ through $e = 2$, in contrast to the CS-DiD null. As discussed in §5.2, we interpret this divergence as extrapolation bias given the degenerate cohort structure and short post-period, and we lead with CS-DiD.

A5. Goodman-Bacon decomposition weights

2×2 comparison	Estimate	Weight share	Type
2023 vs never	−0.0755	83.96%	Clean
2024 vs never	−0.0040	1.37%	Clean
2025 vs never	−0.1710	0.68%	Clean
2023 vs 2024 (pre)	−0.0767	2.33%	Clean
2024 vs already-treated 2023	+0.0682	4.66%	Contaminated
2023 vs 2025 (pre)	+0.0399	4.66%	Clean
2025 vs already-treated 2023	−0.0325	2.33%	Contaminated
2024 vs 2025 (pre)	0	0.00%	Clean
2025 vs already-treated 2024	0	0.00%	Contaminated

Source: `analysis/robustness/bacon_operating_margin_win.csv`. 92.99% of identifying weight comes from clean (treated-vs-never or treated-vs-not-yet) comparisons; 7.00% from contaminated (already-treated-as-control) comparisons. TWFE contamination is therefore small, and the BJS-CS-DiD divergence cannot be attributed to TWFE negative-weight artifacts.

A6. Net-of-tax sensitivity: full grid

See main-text Table 4 for the net-of-tax sensitivity of the implied per-$B dose slope. Additional details in `analysis/robustness/nettax_sensitivity_operating_margin_win.csv`.

Recapture rate	Slope per $B	Slope SE	Overall ATT	ATT SE	95% CI
0.1000	0.000584	0.000828	+0.00288	0.00408	(−0.00513, +0.01088)
0.1625	0.000628	0.000890	+0.00288	0.00408	(−0.00513, +0.01088)
0.2250	0.000678	0.000962	+0.00288	0.00408	(−0.00513, +0.01088)
0.2875	0.000738	0.001046	+0.00288	0.00408	(−0.00513, +0.01088)
0.3500	0.000809	0.001147	+0.00288	0.00408	(−0.00513, +0.01088)

State-level first-difference regression of Δoperating-margin on per-state net-of-tax dose (in B), ateachof fiverecapture–ratepriors.TheoverallATTisinvariantacrosspriorsbecausethedosescalingenterssymmetricallyonbothsidesof theregression; whatu effect). The confidence interval for the overall ATT straddles zero across the full span, confirming the null is robust to the Baicker-Staiger recapture channel within plausible state-level priors.

A7. Placebo event studies

A7.1 Overall placebo ATTs

Outcome	ATT	SE	95% CI
Total operating revenue ($)	+$16,658,466	$5,434,461	(+$6,006,923, +$27,310,009)
Non-Medicaid IP days	+545	205	(+144, +947)

Source: `analysis/robustness/placebo_outcomes.csv`.

A7.2 Placebo event-time dynamics

See `analysis/robustness/placebo_eventstudy_total_op_revenue.csv` and `analysis/robustness/placebo_eventstudy` for event-time-specific placebo coefficients. The positive post-period pattern in both placebos is consistent with a general-volume channel (§5.4 of main text).

A8. Alternative specifications

See `analysis/robustness/alt_specifications_operating_margin_win.csv` for the full grid. Summary from §5.6:

Specification	ATT	SE	95% CI
5th/95th winsorization	−0.005	0.008	(−0.021, +0.011)
Alternative bandwidth 2018–2025	−0.007	0.009	(−0.025, +0.011)
Never-treated controls only	−0.008	0.010	(−0.028, +0.012)
Inpatient-hospital-only cohort	+0.007	0.005	(−0.003, +0.017)

No specification moves the headline outside (−0.015, +0.010). The inpatient-hospital-only cohort direction flips from negative to positive, consistent with the Table 3 class-specific heterogeneity result.

A9. §71116 hand-coding notes

A9.1 Scope and parse rate

- 26 arrangements meet all three §71116 in-scope criteria: (a) provider class ⍰ {inpatient hospital, outpatient hospital, nursing facility, AMC professional}, (b) rating period starts on or after 2025-07-04, (c) not automatically grandfathered by the approval-date rule.
- 9 of 26 arrangements had parseable Addendum Table 2.A data.
- 17 of 26 arrangements could not be parsed: 14 have no Table 2.A heading in the PDF, 3 have the heading but blank row fields.
- 21 of 26 CMS approval letters explicitly designate the arrangement as “likely qualifying for the temporary grandfathering period in section 71116(b).”

A9.2 Parsed dose distribution (9 arrangements)

Statistic	Value
Mean pre-cap rate (% of Medicare)	80.5%
Median pre-cap rate	80.0%
Max pre-cap rate	101.3%
All observed dose values	0

All 9 parseable arrangements have total (base + SDP) payment at or below the applicable cap (100% of Medicare in expansion states, 110% in non-expansion states). No arrangement in the public corpus has a positive binding §71116 dose.

A9.3 Why not parseable

The 17 unparseable arrangements are not OCR failures. The PDFs are text-extractable (mean 40,000 characters per document). The Excel Addendum Table 2.A is simply not merged into the approval-package PDF by the state. This is a structural feature of CMS public-posting practice, not a technical limitation. A CMS FOIA request for the unredacted Excel workbooks is pending (`data/foia-request-letter-cms.md`).

A9.4 Net-of-tax Task B results

Top-200 arrangements by gross TC (95% of dollar exposure):

- 4 of 200 have populated Addendum Table 4.A (IGT transferring entities with dollar amounts): 3 Louisiana inpatient-and-outpatient hospital arrangements and 1 Florida inpatient-and-outpatient hospital arrangement. Arrangement-specific coverage: $5.8B of $296B gross (2.0%).
- 0 of 200 have populated Addendum Table 5.A (provider tax).
- 196 of 200 retain state-level MACPAC/KFF prior, tagged `state_prior_top200_reviewed` for audit trail.
- 835 of 1,038 bottom arrangements retain state-level prior (tagged `state_prior`).

A9.5 State-prior table (from **`data/scripts/handcoding/build_net_of_tax_handcoded.py`**)

State-level recapture priors compiled from MACPAC October 2024 and KFF syntheses, range 0.10 (Alaska) to 0.35 (Illinois, Michigan). These are deliberate approximations; the §5.5 sensitivity analysis in the main text spans the 10%–35% range to bracket plausible priors.

A10. Data-quality notes

A10.1 Three HTML-href parse errors

3 of 1,038 preprints (~0.3%) have a minor field-parse error (`payment_type = "No state"`) traceable to CMS listing-page HTML quirks. These records are retained in the panel but flagged in analysis.

A10.2 Boilerplate saturation in text-level flags

The CMS preprint template contains near-universal references to "provider tax," "IGT," "100% of Medicare," "80% of Medicare" (baseline FFS benchmark), and "hold harmless." Consequently, text-level flags `has_provider_tax`, `has_igt`, and `has_hold_harmless_lang` fire on > 99% of preprints and carry no discriminating power. We rely on them only as input features for the recapture-rate estimator in `data/scripts/03_build_net_of_tax_treatment.py`, not as stand-alone arrangement-level treatment variables.

A10.3 HCRIS coordination

The acute-hospital HCRIS panel (42,804 hospital-years) is shared with the sibling paper `papers/dsh-liur-threshold-rd/`. We avoid duplicate downloads by reading from the sibling paper's Phase 3 clean output. Any future changes to that panel must be reflected in both papers' Phase 4 analyses.

A11. Reproducibility

Full reproduction from clean data:

```
cd /Users/jonpalisoc/research-pipeline/papers/state-directed-payments-panel
bash analysis/run_all.sh
```

This runs: 1. analysis/main.py — CS-DiD main analysis (Agent 4A) 2. analysis/robustness/robustness.py — robustness battery (Agent 4B)

All outputs land in analysis/tables/*.csv, analysis/figures/*.png, analysis/robustness/*.csv, and analysis/log/*.log. Required packages: pandas >= 2.0, numpy >= 1.24, matplotlib >= 3.7, pyfixest >= 0.30, statsmodels >= 0.14, scipy >= 1.10. No R or Stata required.

A12. Correspondence to main-text tables and figures

Main-text reference	Source CSV	Source figure
Table 1	data/summary-statistics.md, data/clean/sdp_panel.parquet	—
Table 2	analysis/tables/csdid_overall_summary.csv	—
Table 3	analysis/tables/csdid_heterogeneity_operating_margin_win.csv	analysis/figures/fig08_heterogeneity_opera
Table 4	analysis/robustness/nettax_sensitivity_operating_margin_win.csv	analysis/figures/fig07_nettax_forest_opera
Table 5	analysis/robustness/placebo_outcomes.csv	analysis/figures/fig06_placebo_panel_opera
Table 6	analysis/robustness/robustness_summary.csv, analysis/robustness/honestdid_operating_margin_win.csv	analysis/figures/fig04_honestdid_operating
Figure 1	—	analysis/figures/fig01_raw_trends_<outcome (4 panels)
Figure 2	analysis/tables/csdid_eventstudy_*.csv	analysis/figures/fig02_eventstudy_<outcome (4 panels)
Figure 3	analysis/tables/state_treatment_cohorts.csv	analysis/figures/fig03_cohort_support.png
Figure 9	data/clean/sdp_section71116_dose.csv, hand-coding outputs	analysis/figures/fig09_s71116_descriptive.